WeBugMate – AI Powerd Resolution Platform

23SE02IML116 - MAVANI KRISH

MENTOR: Mr. PARTH NATHANI

Abstract

During my internship at We3vision, I worked on four AI-based projects: **Skill Matching Engine, WhatsApp AI Automation, Prompt-Man, and We3vision AI Chatbot**. These projects focused on employee-task matching, WhatsApp response automation, prompt engineering, and multilingual company-specific chatbot development. Using technologies such as FastAPI, n8n, fine-tuned BGE embeddings, and Retrieval-Augmented Generation (RAG), I gained practical experience in artificial intelligence, deep learning, and application development. The projects helped me understand AI model integration, intelligent workflow design, and the development of efficient solutions for real-world business requirements.

Introduction

Artificial Intelligence (AI) plays an important role in developing intelligent applications that automate processes and improve productivity. During my internship at We3vision, I worked on four projects: Skill Matching Engine, WhatsApp AI Automation, Prompt-Man, and We3vision AI Chatbot. These projects focused on employee-task matching, automated WhatsApp communication, prompt engineering, and multilingual company-specific chatbot development. Using technologies such as FastAPI, n8n, fine-tuned BGE embeddings, and Retrieval-Augmented Generation (RAG), I explored practical applications of AI, natural language processing, and automation. The internship provided valuable experience in developing AI-powered solutions, integrating different technologies, and addressing real-world business requirements. It also improved my understanding of software development workflows and the importance of designing reliable, user-focused applications. This experience helped me connect academic knowledge with practical industry requirements.

Materials

- Programming Languages: Python, JavaScript, TypeScript
- Backend Framework: FastAPI
- Frontend Technologies: Next.js, React, Tailwind CSS
- AI/ML Technologies: Fine-tuned BGE Embeddings, Generative AI, NLP
- AI Integration: OpenRouter API
- Automation: n8n, WhatsApp Business API
- RAG & Information Retrieval: ChromaDB, Embedding Models
- Database & Storage: Supabase, PostgreSQL, Google Sheets

Methodology

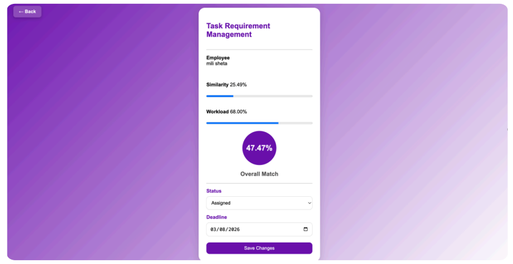
The methodology followed during the development of the four projects focused on understanding requirements, developing AI-based features, integrating services, and validating system performance.

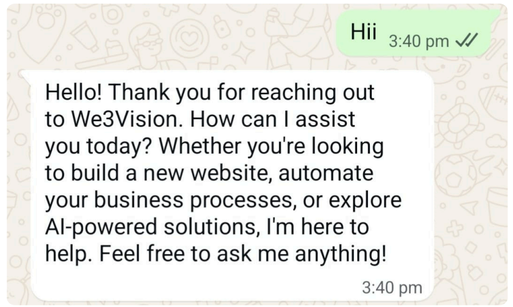
- Requirement Analysis: Defined the objectives and functional requirements of each project.
- System Design: Designed workflows for task matching, WhatsApp automation, prompt enhancement, and chatbot interactions.
- AI Implementation: Applied fine-tuned BGE embeddings for skill matching and generative AI for prompt improvement and chatbot responses.
- Workflow Automation: Integrated n8n with WhatsApp to support automated communication and approval workflows.
- RAG Implementation: Retrieved relevant information from company documents to generate context-specific chatbot responses.
- Testing and Integration: Checked functionality, API communication, response quality, and overall system behavior.

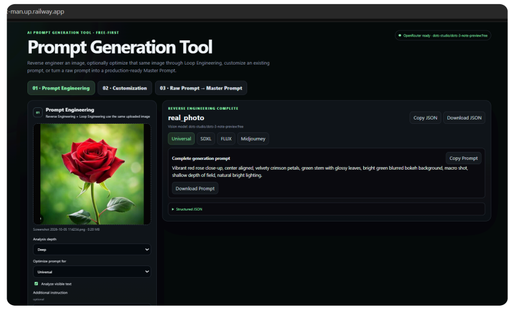
An iterative development process was used to implement and refine project features. This approach supported the development of intelligent applications

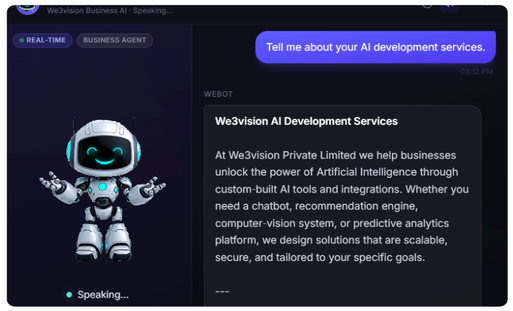
Results

All four projects demonstrated AI-driven automation and intelligent application development.









Conclusion

The internship at We3vision enhanced my knowledge of AI-based task matching, WhatsApp automation, prompt engineering, and multilingual chatbot development. It improved my problem-solving skills and ability to develop practical AI applications. Working on these projects provided experience with generative AI, workflow automation, and information retrieval. It also strengthened my understanding of API integration and application development. Overall, the internship contributed to my technical growth and professional development while building my confidence in developing intelligent software solutions.

Recommendations

The following recommendations can improve the effectiveness and reliability of the Skill Matching Engine, WhatsApp AI Automation, Prompt-Man, and We3vision AI Chatbot.

- Improve AI Responses: Refine model outputs for greater accuracy and relevance.
- Enhance Automation: Expand workflow automation to reduce manual effort.
- Optimize Prompt Quality: Improve prompt refinement and customization features.
- Expand Language Support: Enhance multilingual response quality in the chatbot.
- Regular Maintenance: Update dependencies, test features, and apply security improvements.

Acknowledgements

I sincerely thank We3vision for providing me with the opportunity to work on AI-based projects. I am grateful to my mentors and team members for their guidance, support, and encouragement throughout my internship.